

Supplementary Material for “Layout-Aware Text Column Detection and Reading-Order Recovery for Traditional Mongolian Historical Archives”

Lanying Liang¹ and Yuefeng Liu^{2*}

¹Archives, Inner Mongolia University of Science and Technology,
Baotou, 014010, Inner Mongolia, China.

^{2*}School of Digital and Intelligent Industry (School of Cyber Science
and Technology), Inner Mongolia University of Science and Technology,
Baotou, 014010, Inner Mongolia, China.

*Corresponding author(s). E-mail(s): liuyuefeng@imust.edu.cn;
Contributing authors: 389647973@qq.com;

This supplementary material contains diagnostic experiments moved from the main text to keep the Journal of King Saud University - Computer and Information Sciences submission focused on the main benchmark, core detector comparison, strict page-level metrics, and limitations.

1 Inter-Annotator Agreement

2 Training and Threshold Protocol

3 Supplementary Baselines and Diagnostics

4 Ablation and Runtime

Table 1 Inter-annotator agreement on the 5-page independent subset.

Metric	Value
Box precision@0.5	0.929
Box recall@0.5	0.788
Box F1@0.5	0.852
Mean matched IoU	0.800
Reading-order pairwise agreement	0.677
Ignore F1@0.5	0.286

Table 2 Training-budget and threshold-selection protocol.

Model	Init. domain	Epochs	Img. size	Threshold source
YOLOv8n	COCO	50	960	val F1, th=0.35
Faster R-CNN	COCO	5	960 max-side	val F1, th=0.80
RT-DETR-L	general detection	5	640	val F1, th=0.10
DocLayout-YOLO	DocStructBench	5	1024	val F1, th=0.03

Table 3 YOLOv8n validation threshold sensitivity. The final threshold is selected on validation data only.

Threshold	Val Prec.	Val Rec.	Val F1
0.25	0.746	0.787	0.766
0.30	0.812	0.759	0.785
0.35	0.876	0.722	0.792
0.40	0.873	0.639	0.738

Table 4 Transfer-detector baselines under validation-selected thresholds.

Method	Prec.	Rec.	F1	Mean IoU	RO Acc.
RT-DETR-L + rotation-aware order	0.032	0.270	0.058	0.632	0.847
DocLayout-YOLO + rotation-aware order	0.046	0.031	0.037	0.572	0.000

Table 5 Strict-localization comparison at IoU@0.75.

Method	Prec.	Rec.	F1	Mean IoU	RO Acc.
Heuristic column extraction	0.322	0.447	0.375	0.901	1.000
YOLOv8n + rotation-aware order	0.579	0.434	0.496	0.826	1.000
Faster R-CNN MobileNet + rotation-aware order	0.352	0.289	0.318	0.828	1.000

Table 6 YOLOv8n project-level test performance with different numbers of annotated training pages.

Training pages	Test Prec.	Test Rec.	Test F1	Mean IoU	RO Acc.
7	0.843	0.388	0.532	0.719	1.000
20	0.875	0.691	0.772	0.748	0.900
43	0.956	0.717	0.820	0.770	0.900

Table 7 Diagnostic column-aware YOLO post-processing.

Setting	Split	Prec.	Rec.	F1	Mean IoU	RO Acc.
Main YOLO threshold 0.35	test	0.956	0.717	0.820	0.770	0.900
Recall-oriented threshold 0.30	test	0.937	0.776	0.849	0.761	0.818
Column-aware prior filter	val	0.929	0.791	0.854	0.779	0.875
Column-aware prior filter	test	0.949	0.730	0.825	0.760	0.818

Table 8 Ablation of ignore-region filtering and rotation-aware reading order at score threshold 0.35.

Configuration	Prec.	Rec.	F1	RO Acc.	FP/FN	Removed
Full: ignore + rotation-aware order	0.956	0.717	0.820	0.900	5 / 43	0
w/o rotation-aware order	0.956	0.717	0.820	1.000	5 / 43	0
w/o ignore filtering	0.956	0.717	0.820	0.900	5 / 43	0
w/o both	0.956	0.717	0.820	1.000	5 / 43	0

Table 9 Runtime on the 11-page test split under local CPU execution.

Method	Device	Mean s/page	Median s/page
Heuristic column extraction	CPU	1.843	1.729
YOLOv8n + rotation-aware order	CPU	0.193	0.196
Faster R-CNN MobileNet + rotation-aware order	CPU	0.168	0.162